eResearch Australasia 2026

From twenty-dollar guesses to verifiable data

AI-assisted qualitative analysis done properly

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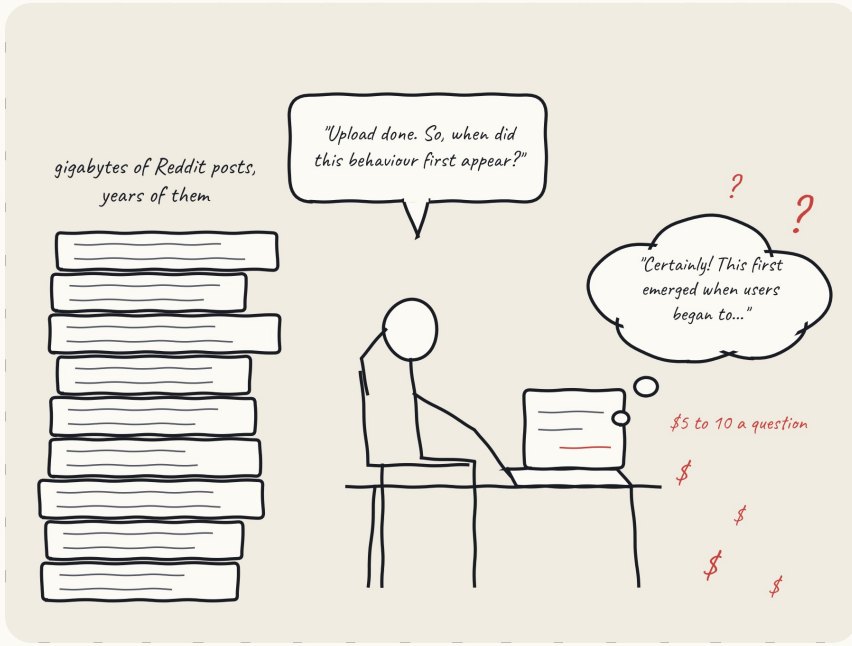
Australian Research Data Commons

Where this story starts

Keen to use AI. Stuck in a chat window.

- Generative AI is everywhere, and researchers want to use it.
- Without programming skills or training, most attempts never get past chatting with it.
- Chat interfaces were never designed for reproducible research.

A psychology PhD student, gigabytes of Reddit posts, starting with a simple question: how did a behaviour emerge over time? Her method: upload it to ChatGPT and ask.



What went wrong for her

Where chat falls short for research

A question without the data

She could not define the question precisely without knowing her data well.

Paying to re-read the data

$5 to $10 a request just to read what she uploaded, often far more. No guarantee on the quality.

Unverifiable

Even a useful answer: correct, biased or hallucinated? No way to check.

Irreproducible

Same prompt, same dollars, no guarantee the model reads the data the same way or answers the same.

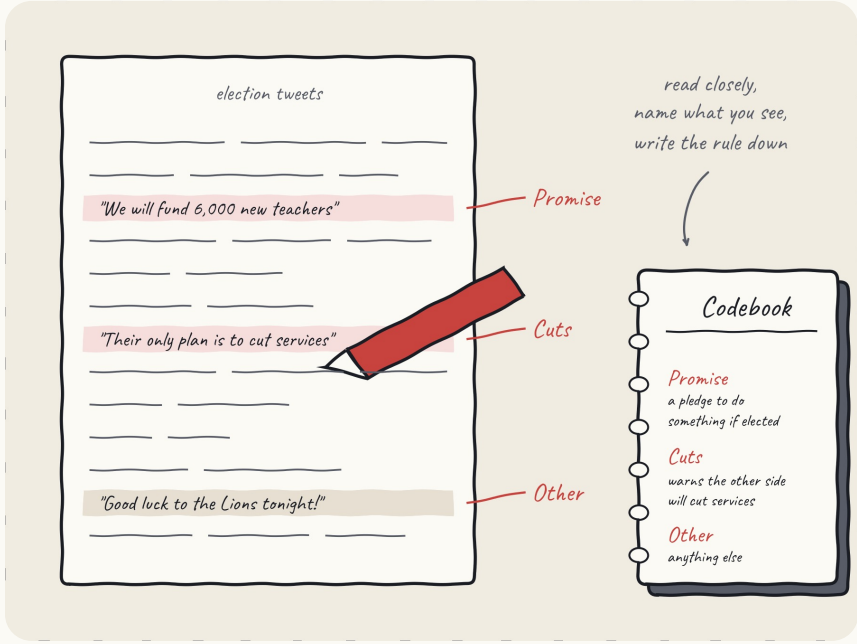
> **HASS researchers need a method and a tool that make generative AI effective and efficient, not just available.**

How qualitative researchers work

Coding is interpretation, written down

- **Read closely.** Researchers read passages and label each with a code.
- **Name what you see.** Codes grow out of the data and the research question.
- **Write the rule down.** A codebook defines every code so others can apply it the same way.

This is not sorting text into preset categories, and not letting a machine find clusters. It is the researcher's judgement, written down so it can be checked and counted.



How qualitative researchers work

Agreement takes several rounds

1 · Code independently

two or more coders, the same sample, without seeing each other's work

→

2 · Measure agreement

Cohen's kappa, Krippendorff's alpha

→

3 · Discuss the disagreements

each one shows a code that is vague or overlaps another

→

4 · Refine and repeat

sharpen the codebook, code a fresh sample, measure again

↻ round after round, until agreement reaches what the field accepts

> **Every round sharpens three things at once: the question, the codebook, and the shared reading of the data.**

This loop is the method. Any new coder, human or AI, has to go through it.

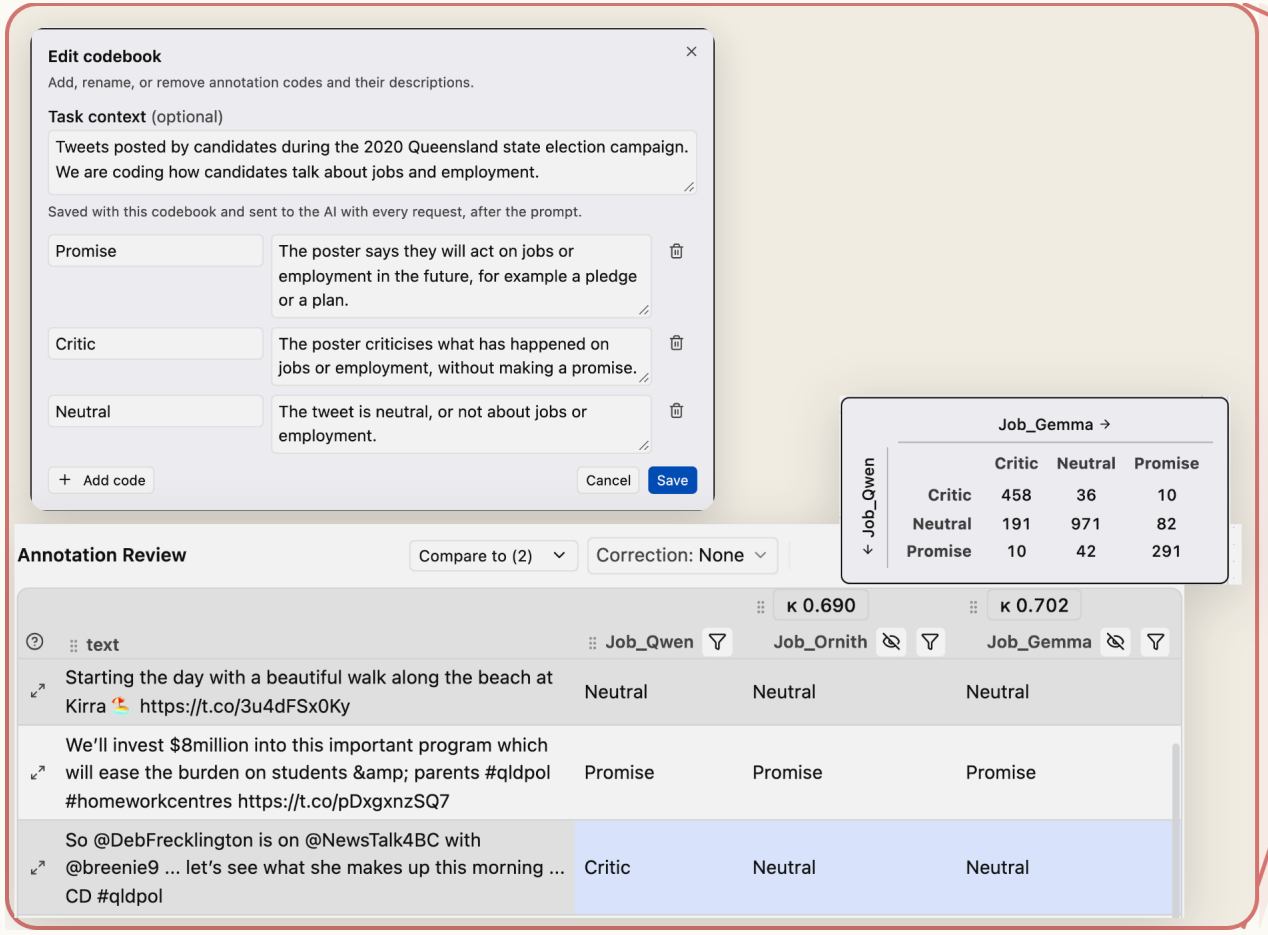
Our premise

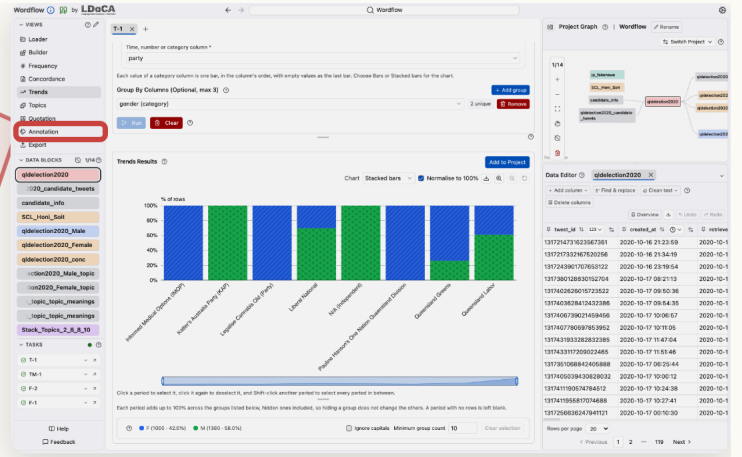
AI should not replace critical thinking in research, and no model deserves blind trust.

So we designed the tool around **what chat gets wrong**, and kept the coding method as it is.

What we built

Wordflow and its Annotation tool





Wordflow is LDaCA's open-source app for text analysis, without programming.

Annotation is one of its tools: our attempt to make responsible AI-assisted coding available to researchers.

The workflow

AI joins an established method

1 · Read and hand-code

researchers learn the data first

2 · Draft a codebook

codes and definitions from that reading

3 · Trial the AI as a coder, and iterate

AI codes a sample → check agreement and the failed cases → refine the codebook, add examples or switch model → repeat

The gate

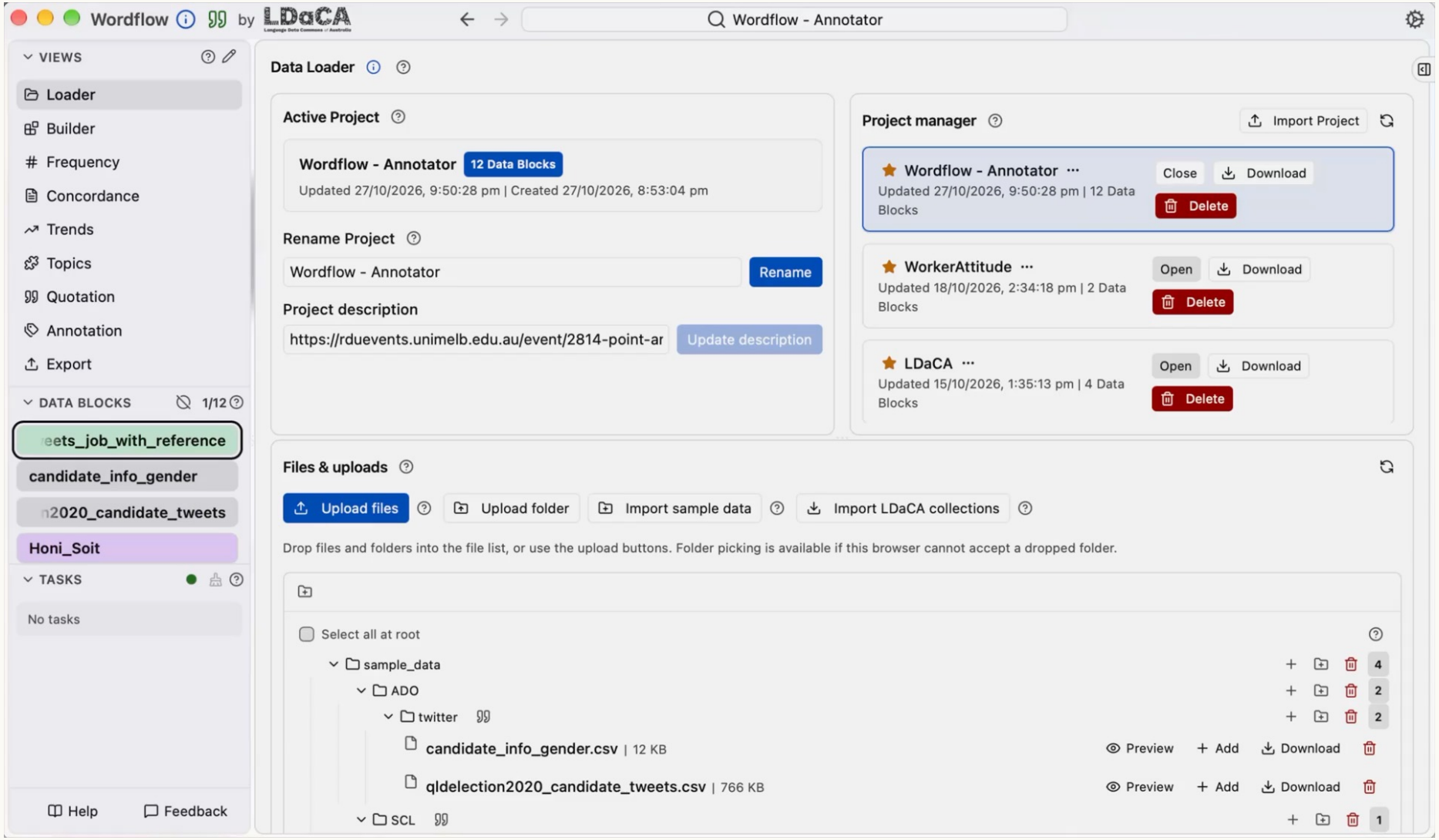
AI–human agreement comparable to human–human. The researcher makes the call.

→

4 · AI scales it up

the refined codebook, across the full corpus

The method stays the same. **The AI is checked like any new coder before it codes at scale.**



Demo

Step 1
Code by hand

Step 2
Trial the AI

Step 3
Scale up

1 min 22 s, no sound

The model's place

A new coder, under the researcher’s rules

What it adds

Scale: thousands of passages in hours

Consistency: the same codebook for every passage

A second opinion: where it disagrees with us, the codebook needs work

A choice of coders: swap the model, not just the codebook, and compare

What keeps it in check

Worked examples: from the researchers' own coding

The same checks: kappa, alpha and a confusion matrix

The same team process: disagreements are argued out and the codebook refined, as between people

Reproducibility

A run that can be described and repeated

A defined run

One template, one model, one set of rows.

A recorded history

The Project graph shows how each Data Block was made.

The codebook is saved too

Codes and definitions are kept and exported with the Project.

Ready for a methods section

Codebook, prompt, model and agreement score: all there.

How Wordflow harnesses the model

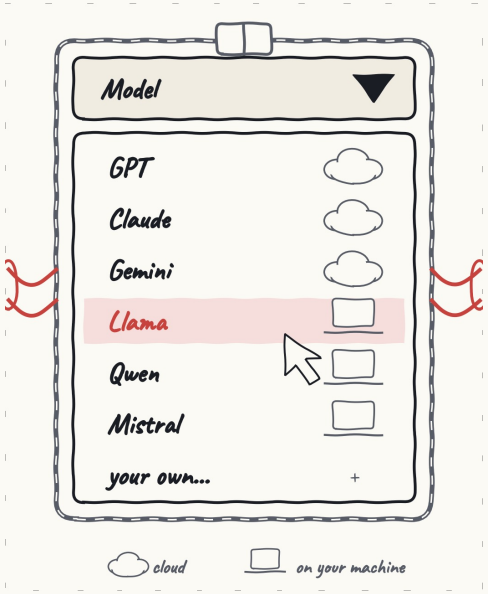
Structured in, structured out

In

The codebook: codes and definitions

Worked examples from human coding

One passage at a time, same template



any model, same harness

Out

One code from the codebook per passage

A column beside the source text

Every answer lands in the same shape, so it can be **checked, compared and further analysed** by other tools, seamlessly within Wordflow.

Choosing a model in Wordflow

Annotator works with all three kinds of provider

Provider	Strengths	Trade-offs
Commercial	fast, most capable, wide choice	cost per use, data leaves the institution, ethics questions
Local	data stays on the machine, open-source models, little or no cost	less capable, slower, needs technical set-up
Institutional	strong privacy and confidentiality guarantees, usually subsidised	still a cost to the researcher; only where the institution provides and supports it

Same codebook, same checks, same output, whichever provider is behind it. **Ethics approval still governs:** the provider changes where the data goes, not what may be done with it.

Accountability

The researcher answers for the results

Deciding what the codes are

The codebook comes from reading the data. That is the researcher's interpretation.

Analysis shaped by the experts

Expertise and deep familiarity with the texts and theory drive the analysis.

Coding that was never checked

Unchecked, AI codes are no more trustworthy than a chat answer. Only faster and tidier.

Vouching for itself

A model cannot tell you its coding is right. Only agreement with human coders can.

AI can boost productivity, stretch imagination and take qualitative analysis to a scale that was out of reach. The researcher is still fully accountable for the results. No model deserves blind trust.

Open source and free

Bring AI into the method. Try it with your researchers.

Online at wordflow.ldaca.edu.au, desktop app or Binder

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Get Wordflow

Thank you

Wordflow is built at the **Sydney Informatics Hub**, The University of Sydney, for the **Language Data Commons of Australia**, a co-investment partnership with the **ARDC**, enabled by NCRIS.

Its cloud services run on the ARDC Nectar Research Cloud.

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